Spatial representations emerge from temporal event credit in a hippocampus-inspired navigation agent without spatial supervision

Can spatial representations emerge from sensory inputs and learning objectives that do not explicitly encode space? We investigate this question using a bottom-up computational architecture inspired by hippocampal physiology, exploring whether spatial organization can arise from the temporal structure of sensory experience and self-generated behavior, without position labels, maps, or spatial rewards.

Our model combines dentate gyrus (DG)-like sparse sensory events with CA3-like sequential memory. Temporal relationships between encountered DG events provide two complementary learning signals: a **"push"** that reinforces DG activity more strongly after longer intervals between events, and a **"pull"** that rewards shorter transitions between sensory events or, in goal-conditioned variants, faster arrival at internally defined DG targets. Together, these pressures aim to make temporal separation a more informative proxy for navigational distance. The agent learns from egocentric visual experience and intrinsic motivations, without explicit information about its physical location or spatial distances.

During exploration of a continuous visual environment, spatially informative DG activity emerges. To isolate the contribution of temporal information, we compare temporally weighted DG feedback against mean-matched constant feedback, evaluating the resulting representations on identical sensory observations. Temporal feedback produces higher DG spatial information, less redundant response maps, and a broader spatial distribution of field peaks across five paired training seeds. These improvements extend to downstream decoder representations, where spatial information also increases consistently. In contrast, temporally weighting the policy's target-hit reward does not consistently improve exploration or spatial representation quality compared with constant hit rewards.

The agent's behavior further shapes spatial coding. Even with network weights frozen, the learned policy produces more spatially informative downstream activity than random actions, demonstrating that spatial coding depends not only on the learned network but also on how the agent interacts with its environment.

Together, our results demonstrate how spatial coding can emerge from the interaction of temporal learning, sequential memory, and self-generated behavior, without explicitly optimizing for spatial information.

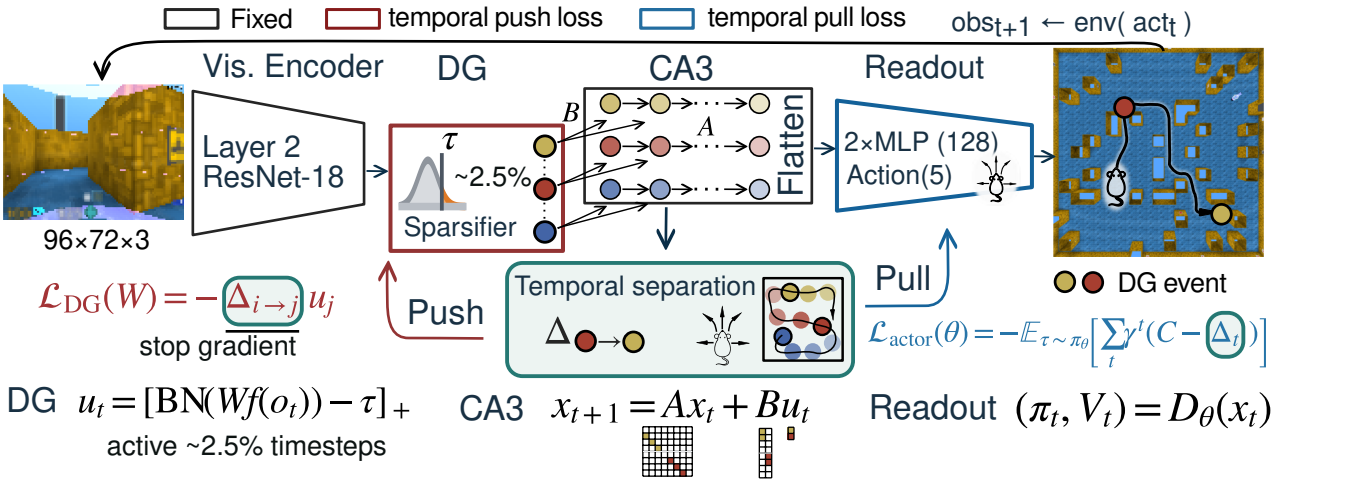


Figure 1: Push–pull architecture. Fixed visual trunk, learned sparse DG, fixed CA3 sequence memory, and policy readout. Red: temporal DG credit; blue: target-hit policy credit.

Spatial emergence and temporal push. DG spatial coding rises during exploration (Fig. 2A). On the same 10,001 observations, temporal DG credit increases the baseline DG spatial score in all five paired seeds (mean +0.0575), reduces map redundancy, and broadens the area covered by eligible field peaks (mean +0.109; Fig. 2C,D). Seed 8 shows the largest peak-area increase among the five pairs. Decoder-1 information also rises by 0.680 bits per activation on average in all five baseline pairs (Fig. 2E). Goal-conditioned comparisons largely point the same way; the frontier condition has only three matched encoder pairs.

Behavior shapes spatial coding. With each trained network frozen, its learned policy elicits more spatially informative decoder-1 activity and more reliable spatial maps than random actions in the matched evaluation (Fig. 2B). This comparison isolates the effect of behavior at evaluation time for a fixed representation and readout; it does not attribute the difference to a change in network weights.

Temporal pull. Replacing temporally weighted target-hit credit with constant hit credit has mixed effects on coverage (Fig. 2F) and DG fields. Both rewards reinforce target arrival, while actor–critic discounting favors earlier arrival. A separate longer-credit intervention changes PPO discount and GAE lambda jointly; its higher frontier-policy coverage and lower spatial information were measured on different policy rollouts, so they do not isolate a fixed-input DG-code change. Assessing goal control requires matched-command and physical-arrival analyses.

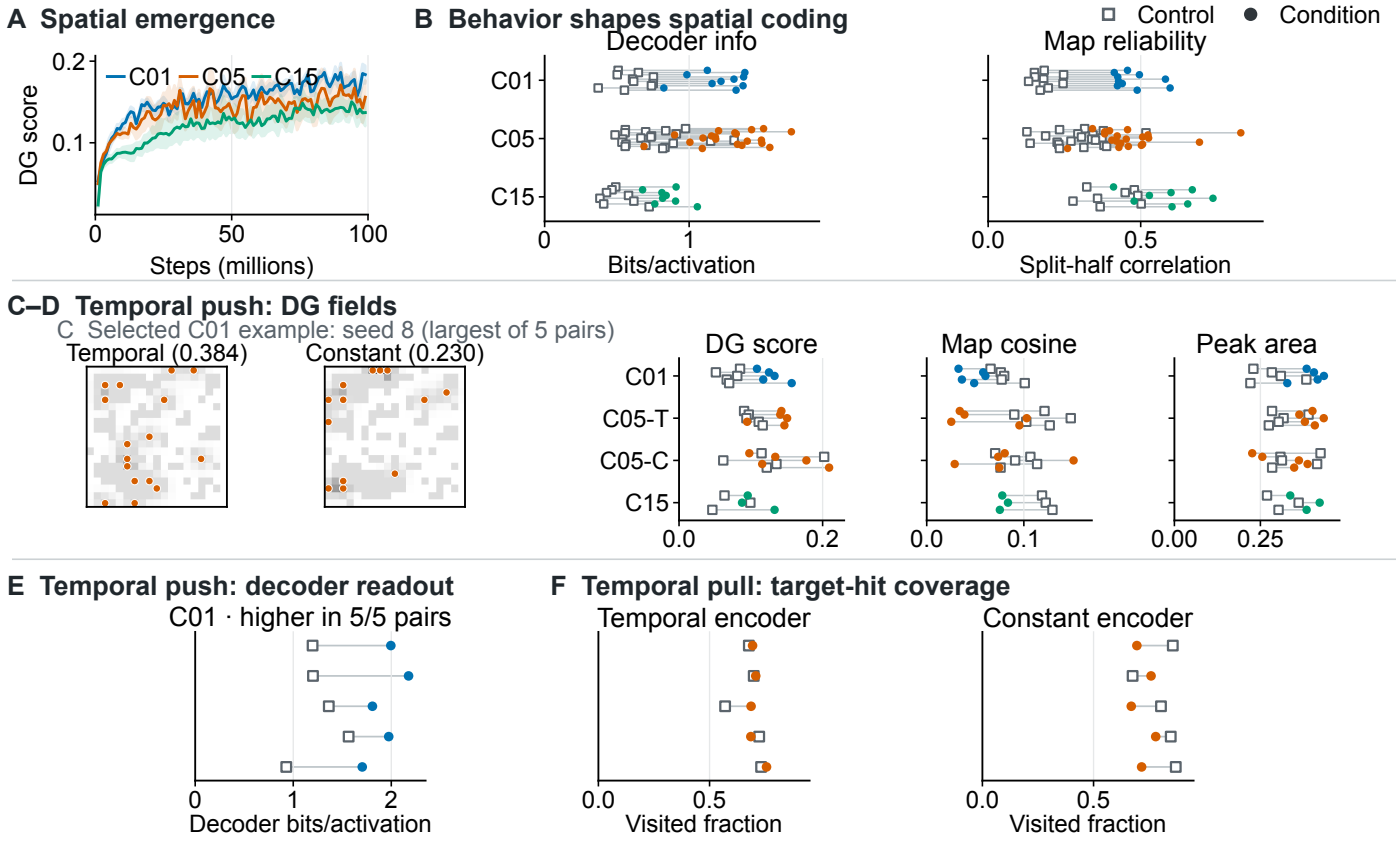


Figure 2: Results grouped by claim. All dot plots show absolute values: gray open squares are controls, filled colored circles are tested conditions, and thin gray connectors pair training seeds. C01 is baseline, C05 is goal-conditioned policy, and C15 adds frontier goal choice. **A,** Online DG spatial score (mean ±1 s.e.m.; five seeds per family). **B,** Random-action control versus each frozen model's own policy: decoder-1 spatial information and split-half map reliability (38 models, with two evaluation repeats averaged per model). **C,** Selected C01 seed 8, chosen for the *largest* temporal-minus-constant peak-area contrast among five paired seeds: 0.384 versus 0.230. Orange dots mark eligible field peaks after a 0.25% activity cutoff on shared observations; gray bins were unvisited. **D,** Constant versus temporal DG credit on identical observations for every available encoder pair: DG spatial score, active-map cosine, and peak area. C01, C05-T, and C05-C have five pairs each; C15 has three. The C05-T/C comparisons hold temporal or constant worker bonus fixed. **E,** C01 decoder-1 information for constant versus temporal DG credit, higher in all five pairs (mean temporal-minus-constant +0.680 bits per activation). **F,** C05 visited fraction with constant-hit control versus temporal-hit worker credit and the encoder fixed (five pairs per encoder arm; mean temporal-minus-constant effects +0.026 and −0.087). This measures coverage; goal control requires the separate matched-command and physical-arrival analyses.